

Subject-Normalized Stress-State Discovery

A baseline-aware representation study using HR, TEMP, and EDA features

Key claim

Baseline removal and subject-level validation are essential before interpreting wearable biosignals as stress states.

Research question

After removing subject baselines, do HR/TEMP/EDA features contain generalizable stress-phase structure, and how strongly do they relate to affect ratings?

Data

312

observations

26

individuals

Phase design

Pre-rest

Puzzle

Post-rest

51 HR/TEMP/EDA features plus 11 affect variables.

Pipeline

01

Median imputation

02

Subject z-score

03

PCA + KMeans

04

CCA + embeddings

Model Checks

0.075

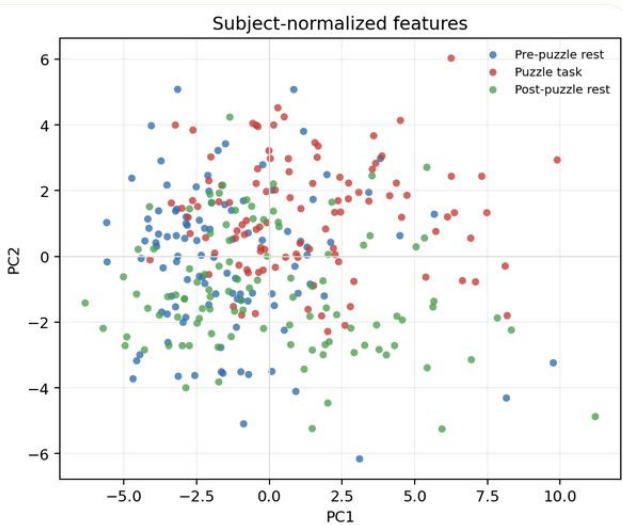
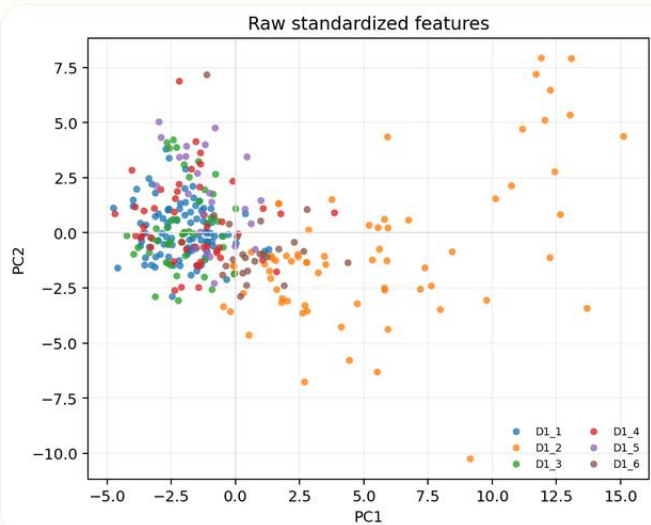
KMeans ARI

0.779

LOSO AUC

Unsupervised structure is weak, but a supervised subject-held-out sanity check confirms nonzero stress-related signal.

1 PCA: Baselines Dominate Low-Dimensional Structure



0.030

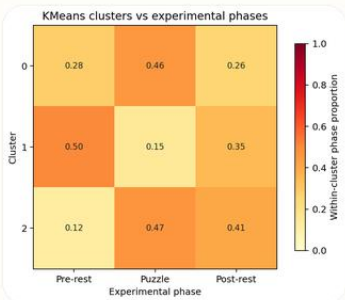
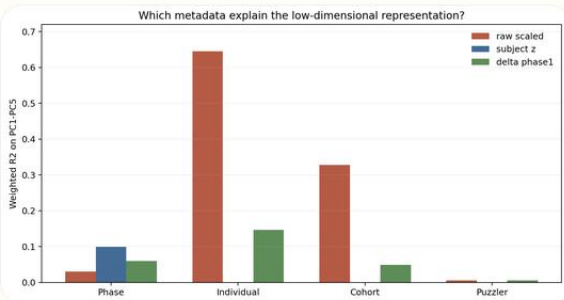
Raw phase R2 in PC1-PC5

0.645

Raw individual R2 in PC1-PC5

0.075

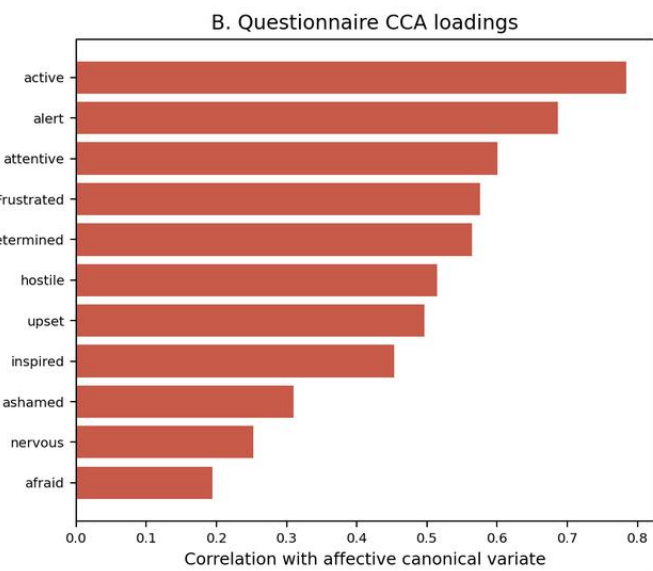
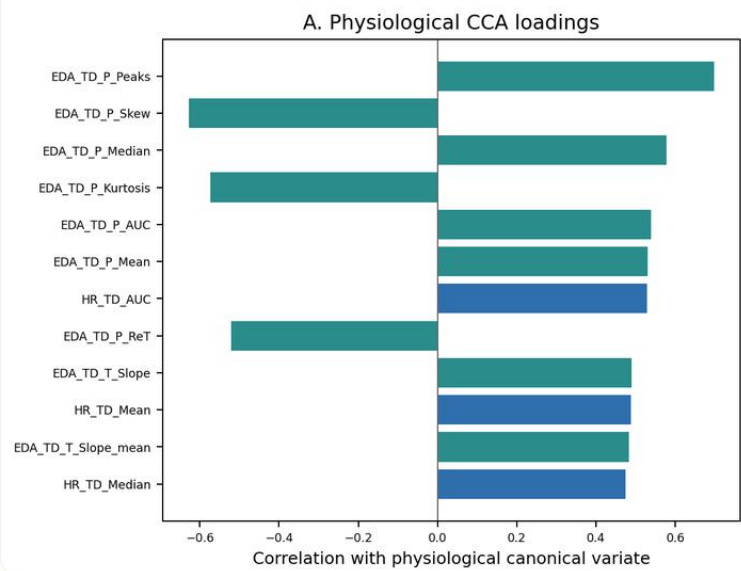
Best KMeans ARI after subject z-score



- Subject and cohort effects dominate raw PCA; phase explains little variance.
- Subject normalization improves phase signal, but KMeans still gives mixed phase clusters.
- Conclusion: use PCA as a diagnostic representation, not as proof of clean stress-state discovery.

2 CCA: Cross-Modal Physiology-Affect Coupling

CCA: physiology-affect coupling, in-sample $r=0.477$, GroupKFold $r=0.362 \pm 0.128$



0.477

In-sample canonical correlation

0.362

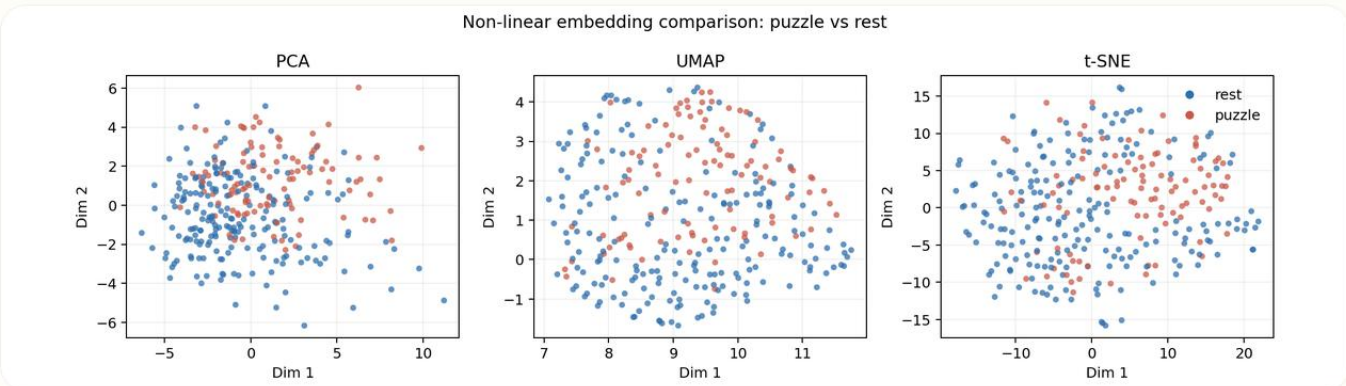
Held-out-subject GroupKFold r

p=0.005

Subject-block permutation test

EDA-phasic variables are strongest; affective loadings form an arousal-like axis.

3 PCA vs UMAP vs t-SNE: Non-linear Comparison



Method	Sil.	kNN acc.
PCA	0.107	0.737
UMAP	0.069	0.769
t-SNE	0.058	0.763

UMAP has slightly higher kNN accuracy, but lower silhouette. Non-linear embeddings do not clearly improve phase geometry over interpretable PCA.

Key Take-aways

1. Individual/cohort baselines dominate raw wearable features.
2. EDA-phasic activity is the strongest cross-modal signal carrier.
3. CCA supports moderate arousal-like physiology-affect coupling.
4. No robust unsupervised separation of emotional states.